UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

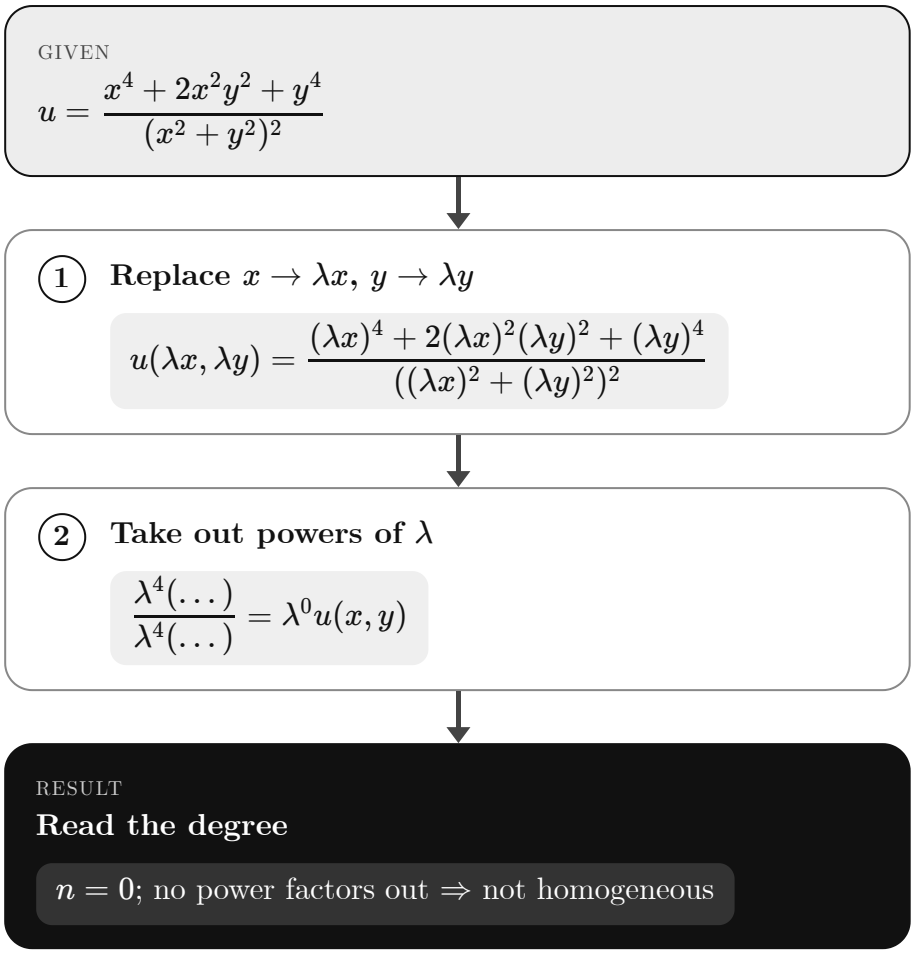
Class 006 · Problems on Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

Contents

1. Figure F1.1 · The scaling test
2. Figure F3.1 · Which Euler formula?
3. Figure F4.1 · Using the extension (slides 006·16–17)
4. The extension for each outer function

1 Testing homogeneity: the scaling test

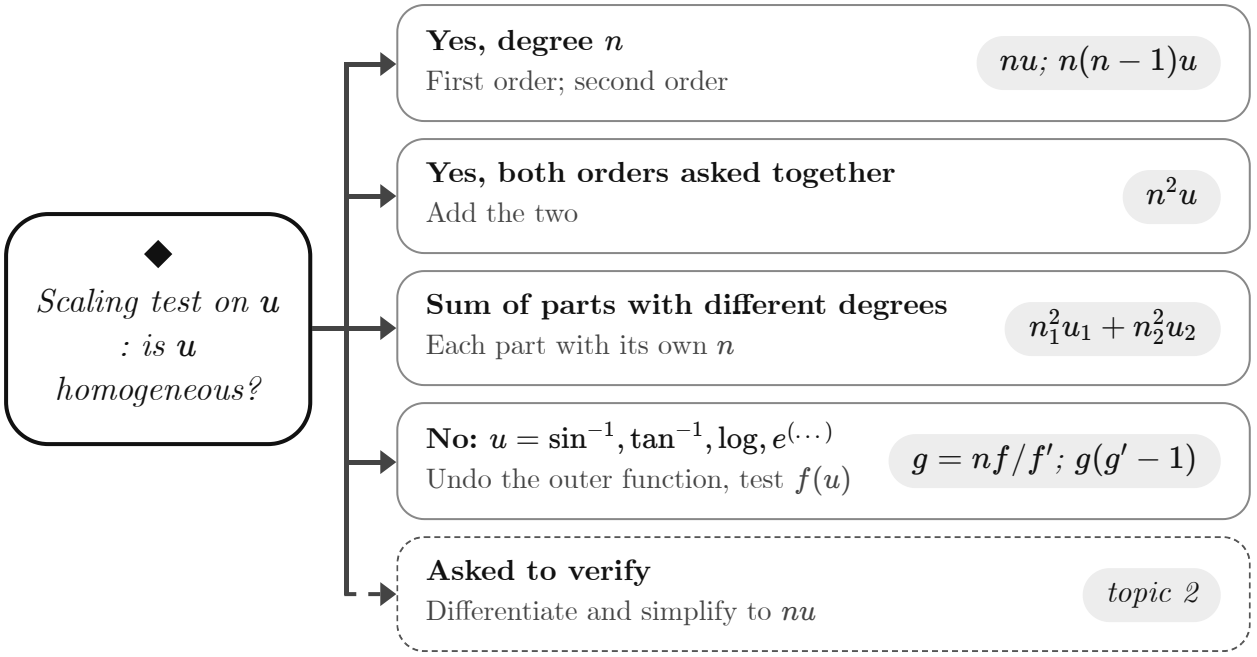


Remark. Use the test on u (topic 3) or on $f(u)$ (topic 4); the steps are the same.

Figure F1.1. The scaling test ESA ★★

S006·5 · T006b 00:01:41 · T006b 00:02:37

3 When u itself is homogeneous: first and second order

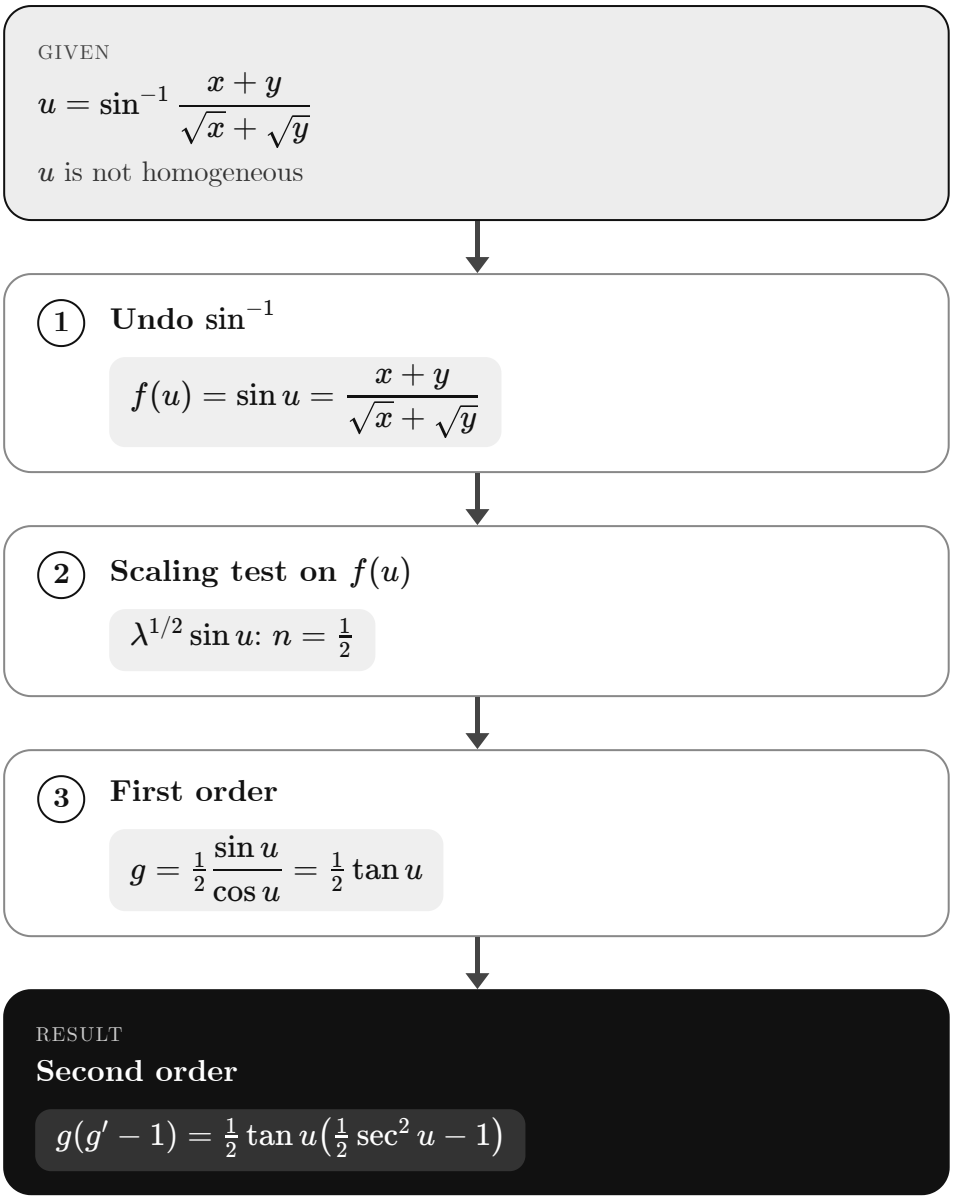


Remark. Exam questions say “using Euler's theorem”: always state the homogeneity and the degree first.

Figure F3.1. Which Euler formula? ESA ★★★

T006b 00:06:11 · T006b 00:09:08 · S006·13

4 When f(u) is homogeneous: the extension



Remark. The same four steps give $4u\log u$ for $e^{(\cdots)}$ (slide 006·12) and $1, -1$ for $\log(\dots)$ (slide 006·14).

Figure F4.1. Using the extension (slides 006·16–17) ESA ★★★

S006·16 · S006·17 · T006b 00:11:40

The extension for each outer function

$u =$	$f(u)$	$g(u) = xu_x + yu_y$	$g(g' - 1)$	**In class 006**
$\sin^{-1}(P)$	$\sin u$	$n \tan u$	$n \tan u \, (n \sec^2 u - 1)$	$n = -3$ (slide 4); $n = \frac{1}{2}$ (slide 16)
$\cos^{-1}(P)$	$\cos u$	$-n \cot u$	$-n \cot u \, (n \csc^2 u - 1)$	$n = \frac{1}{2}$ (slide 3, u itself)
$\tan^{-1}(P)$	$\tan u$	$\frac{n}{2} \sin 2u$	$\frac{n}{2} \sin 2u \, (n \cos 2u - 1)$	$n = 2$ (ESA May/June 2024)
$\log(P)$	e^u	n	$-n$	$n = 1$ (slide 14)
e^P	$\log u$	$nu \log u$	$nu \log u \, (n \log u + n - 1)$	$n = 4$ (slide 12)

ADDED FOR CLARITY S006·4 · S006·13 · S006·15 · S006·17 · ESA May-24 1a